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LAB : MODEL COMPARISON DASHBOARD

Build a complete model selection pipeline: load data, preprocess, train 4 models, evaluate with cross-validation, tune hyperparameters, and produce a final comparison report. We use the Pima Indians Diabetes dataset as the benchmark.

Step 1 — Load, Clean, Split

```
# ■ Load and Clean ■
import pandas as pd, numpy as np, matplotlib.pyplot as plt, seaborn as sns
from sklearn.model_selection import StratifiedKFold, cross_validate, GridSearchCV
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import classification_report, confusion_matrix, ConfusionMatrixDisplay
import time, warnings; warnings.filterwarnings('ignore')

# Check if 'diabetes.csv' exists, if not, download it
import os
if not os.path.exists('diabetes.csv'):
    # Using wget to download the file from a common public repository
    !wget https://raw.githubusercontent.com/plotly/datasets/master/diabetes.csv

df = pd.read_csv('diabetes.csv')
for col in ['Glucose', 'BloodPressure', 'SkinThickness', 'Insulin', 'BMI']:
    df[col] = df[col].replace(0, df[col].median())
X = df.drop('Outcome', axis=1)
y = df['Outcome']
print(f'Dataset: {X.shape[0]} rows, {X.shape[1]} features')
print(f'Class balance: {y.value_counts().to_dict()}')
# Output: {0: 500, 1: 268} - slightly imbalanced
```

```
--2026-06-05 01:28:22-- https://raw.githubusercontent.com/plotly/datasets/master/diabetes.csv
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.108.133, 185.199.109.133, 185.199.110.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.108.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 23873 (23K) [text/plain]
Saving to: 'diabetes.csv'

diabetes.csv      100%[=====>] 23.31K  --.-KB/s   in 0s

2026-06-05 01:28:22 (81.2 MB/s) - 'diabetes.csv' saved [23873/23873]

Dataset: 768 rows, 8 features
Class balance: {0: 500, 1: 268}
```

Step 2 — Build Models & Run Cross-Validation

```
# ■ Cross-Validation Leaderboard ■
# Models - LR and KNN need scaling; trees don't (Pipeline handles this cleanly)
models = {
    'Logistic Regression': Pipeline([('sc', StandardScaler()),
    ('m', LogisticRegression(max_iter=1000, random_state=42))]),
    'Decision Tree' : DecisionTreeClassifier(max_depth=5, random_state=42),
    'Random Forest' : RandomForestClassifier(n_estimators=100, random_state=42),
    'KNN' : Pipeline([('sc', StandardScaler()),
    ('m', KNeighborsClassifier(n_neighbors=7))]),
}
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
metrics = ['accuracy', 'f1_macro', 'roc_auc', 'precision_macro', 'recall_macro']
records = []
for name, model in models.items():
    t0 = time.time()
    res = cross_validate(model, X, y, cv=cv, scoring=metrics)
    elapsed = time.time()-t0
    row = {'Model': name, 'Time(s)': round(elapsed, 2)}
    for m in metrics:
        row[m.replace('_macro', '').title()] = round(res[f'test_{m}'].mean(), 4)
        row[f'{m.replace('_macro', '').title()}_std'] = round(res[f'test_{m}'].std(), 4)
    records.append(row)
leaderboard = pd.DataFrame(records).sort_values('Accuracy', ascending=False)
print(leaderboard[['Model', 'Accuracy', 'F1', 'Roc_Auc', 'Precision', 'Recall', 'Time(s)']].to_string(index=False))
```

	Model	Accuracy	F1	Roc_Auc	Precision	Recall	Time(s)
	Logistic Regression	0.7708	0.7320	0.8372	0.7602	0.7219	0.19
	Random Forest	0.7577	0.7236	0.8235	0.7367	0.7170	2.78
	KNN	0.7435	0.7026	0.7912	0.7206	0.6948	0.14
	Decision Tree	0.7082	0.6682	0.7516	0.6851	0.6719	0.12

Step 3 — Tune the Best Model with GridSearchCV

```
# ■ GridSearch Tuning ■
# From the leaderboard, suppose Random Forest wins
# Tune its key hyperparameters
param_grid = {
    'n_estimators': [50, 100, 200],
    'max_depth': [None, 5, 10],
    'min_samples_leaf': [1, 3, 5],
}
# 3x3x3 = 27 combos x 5-fold = 135 fits — runs in ~1 minute
gs = GridSearchCV(
    RandomForestClassifier(random_state=42),
    param_grid, cv=cv, scoring='accuracy',
    n_jobs=-1, verbose=0
)
gs.fit(X, y)
print(f'Best params : {gs.best_params_}')
print(f'Best CV acc : {gs.best_score_:.4f}')
# Compare tuned vs default
default_rf = RandomForestClassifier(n_estimators=100, random_state=42)
default_scores = cross_val_score(default_rf, X, y, cv=cv, scoring='accuracy')
tuned_scores = cross_val_score(gs.best_estimator_, X, y, cv=cv, scoring='accuracy')
print(f'\nDefault RF accuracy: {default_scores.mean():.4f} ± {default_scores.std():.4f}')
print(f'Tuned RF accuracy: {tuned_scores.mean():.4f} ± {tuned_scores.std():.4f}')
```

Best params : {'max_depth': 5, 'min_samples_leaf': 3, 'n_estimators': 50}
Best CV acc : 0.7759

Default RF accuracy: 0.7577 ± 0.0266
Tuned RF accuracy: 0.7759 ± 0.0367

Step 4 — Final Evaluation + Confusion Matrix Dashboard

```
# ■ Final Evaluation Dashboard ■
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, confusion_matrix, ConfusionMatrixDisplay, classification_report

# One final hold-out test — we only do this ONCE
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42, stratify=y)
fig, axes = plt.subplots(1, 4, figsize=(18, 4.5))
Final_models = {
    'Logistic Regression': Pipeline([('sc', StandardScaler()), ('m', LogisticRegression(max_iter=1000))]),
    'Decision Tree': DecisionTreeClassifier(max_depth=5, random_state=42),
    'Random Forest': gs.best_estimator_, # Tuned version
    'KNN': Pipeline([('sc', StandardScaler()), ('m', KNeighborsClassifier(n_neighbors=7))]),
}
final_results = []
for ax, (name, model) in zip(axes, Final_models.items()):
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    acc = accuracy_score(y_test, y_pred)
    cm = confusion_matrix(y_test, y_pred)
    ConfusionMatrixDisplay(cm, display_labels=['Healthy', 'Diabetic']).plot(
        ax=ax, cmap='Blues', colorbar=False)
    ax.set_title(f'{name}\nAcc: {acc:.2%}', fontsize=9)
    final_results.append({'Model': name, 'Test Accuracy': f'{acc:.2%}'})
plt.suptitle('Final Confusion Matrices — All Models', fontsize=13, y=1.02)
plt.tight_layout(); plt.show()
# Final report for best model
best_model = Final_models['Random Forest']
print('\n=== FINAL REPORT: Random Forest (Tuned) ===')
print(classification_report(y_test, best_model.predict(X_test),
    target_names=['Healthy', 'Diabetic']))
```

